

Ethan Liu

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Technical Skills

Skills & Tools: Python, Swift, Java, JavaScript/TypeScript, C/C++, Bash, SQL (PostgreSQL, MySQL), Go, React.js, Next.js, Tailwind, Node.js, Flask, FastAPI, Vite, Swagger, TensorFlow, OpenCV, GPT, Claude, AWS (S3, EC2, Route 53), GitHub Actions, Docker, Nginx, Git, Splunk, Selenium, Tableau, Twilio, Tauri, Verilog, VHDL, macOS, iOS, Linux, Windows, Android, Chrome, UNIX, Apple Tools (PR3, Radar, TSTT, Quick, Xcode)

Professional Experience

Software Tools and Automation Engineer - Apple

Aug 2025 – Present

- Migrated Darwin Kernel test coverage from a legacy integration service to a **Kubernetes**-based integration pipeline, building REST endpoints and refactoring static configuration to query upstream services dynamically.
- Designed and shipped an ad-hoc kernel test dispatch system that enabled engineers to manually kick off urgent builds, reducing integration turnaround from hours to minutes.
- Built a new hardware test lab from scratch by provisioning Mac hosts, coordinating static networking, validating remote device restore workflows, and resolving low-level hardware failures.
- Audited kernel test container coverage, removed stale suites, and improved CI signal quality across multiple hardware platforms.
- Built an internal read-only bug context service that exposed triage data to contractors and partner teams without escalated permissions, reducing lookup time and unblocking cross-team workflows.
- Owned migration and maintenance of an internal reminder service, including host migration, DNS updates, and production support with zero downtime.
- Led daily integration standups, trained new team members, and authored onboarding documentation for Darwin Kernel triage and integration workflows.

Software Engineer/Business Analyst - Avere

Feb 2025 – Aug 2025

- Facilitated conversations with stakeholders to identify potential issues and clarify project goals for the DMV integration project
- Collaborated with both technical and business teams to gather and validate system requirements
- Defined and documented new system integration endpoints to support DMV data exchange
- Designed and created data migration tables to ensure smooth integration between legacy and new systems
- Improved backend reliability and frontend user experience by fixing backend issues in Java Spring and implementing updates to the React UI for styling that covered 20% of the system, resulting in smoother user interactions and a more accessible product across dev, UAT, and prod environments.
- Built a full-stack automation bot using Node.js, Puppeteer, and Java to monitor and post Cal eProcure bids daily, resulting in 3× faster bid detection and posting through the addition of parallel processing, deduplication, and robust error handling.
- Increased end-to-end test reliability by maintaining 30+ Selenium test cases and collaborating with QA/dev teams to reproduce and resolve critical failures in < 24 hours resulting in a 3× reduction in test cycle time, accelerating feedback and boosting team productivity.

Founding Engineer - Zinley

Mar 2024 – Aug 2025

- Led **full-stack** development in a fast-paced AI startup, launching to the public.
- Drove backend automation with Supabase, implementing custom endpoints, PostgreSQL infrastructure, and policy enforcement to streamline project creation.
- Developed Go backend for user settings, projects, and **LLM** prompts with failover orchestration.
- Led **SRE**, building a status page (Atlassian) integrated with **Datadog** for automated logging & monitoring, ensuring **99.98%** uptime for both frontend and backend.
- Created SnowX.io and Zinley Documentation (**Next.js**) and Statuspage (Atlassian libraries).
- Built API documentation with OpenAPI/Swagger for seamless developer integration.
- Ranked top **5%** of **27,000+** in YCombinator W24; featured in UC Berkeley AI lecture.
- Produced tutorials/demos (iMovie, CapCut, Canva), growing engagement via social media & KPIs.
- Collected user feedback from Discord & Reddit, driving product improvements.

CoreOS Software Engineer (CE) - Apple

Jun 2023 – Nov 2023

- Developed a **Splunk** query to identify over 63 persistent failures within the system. Reported more than 250 bug reports for software builds.
- Conducted **livability testing, blackbox testing, and triaging** for devices on all Apple platforms, saving hours per test cycle.
- Improved overall testing efficiency by integrating **XCTest** frameworks with **Python automation**, leading to a 30% reduction in test execution cycles.
- Collaborated with cross-functional teams to integrate the Python script with existing monitoring tools, resulting in a 75% improvement in overall pipeline health visibility and proactive issue resolution.

Tech Advisor - Apple

Jun 2022 – Feb 2024

- **Researched** and **resolved** complex technical issues for Apple customers, resulting in a 90% customer satisfaction rate and a 20% decrease in call escalations to senior support.
- **Documented** detailed troubleshooting processes for common Apple product issues, reducing average handling time by 10% and improving overall team efficiency.
- Collaborated with cross-functional teams to implement new communication protocols between tech advisors and engineers, leading to a 15% increase in bug resolution speed.

Fullstack Developer Intern- Ren Energy

Mar 2022 – Aug 2022

- Led a team of interns in developing and deploying a robust data validation API server using **JavaScript**, **TypeScript**, and the **Node.js** framework, resulting in a 40% improvement in data accuracy and efficiency metrics.
- Automated data validation processes for API server responses, leading to a 60% increase in data integrity and a 90% reduction in manual error rates.
- Designed and implemented a new data validation process that boosted database accuracy by 30%.
- Collaborated with a fullstack developer to ensure seamless integration of the data validation API into the existing application stack.

Data Clerk - Purple Cow Agency

July 2020 – Dec 2020

- Structured and organized client data for 200+ customers, utilizing advanced **Excel** functions and **MySQL** databases, resulting in improved data quality.
- Utilized **Tableau** to analyze client data, identifying key trends and correlations that led to a 15% increase in client retention rates.
- Implemented **ELT pipelines** to efficiently process client data, leading to a 30% decrease in data processing time and enabling real-time analytics for improved decision-making.

Software QA Specialist - Finders Tree

Sep 2018 – May 2022

- Executed thorough exploratory and automated testing of SaaS websites using **Selenium** in **Java**, resulting in a 20% decrease in critical bugs.
- Documented and diagnosed bugs found during the testing process, resulting in a 15% reduction in overall bug resolution time and improving the user experience.
- Conducted manual testing on website functionality and security measures, identifying and rectifying 6 critical vulnerabilities.

Embedded Software Engineer Intern - NETWORK Sound

Nov 2019 – May 2020

- Collaborated with hardware engineers to conduct testing on the embedded software, achieving a 98% success rate in connection stability and compatibility with various devices.
- Revamped the **ESP32** coding structure in **C**, leading to a 30% improvement in system performance and a 25% reduction in overall energy consumption.
- Collaborated with cross-functional teams to integrate **CAPWAP** and **IAPP** protocols into the software system, enhancing network performance by 30%.

Project Experience

C.H.O.N.K

Feb 2025 – Present

- Automated data collection using **DuckDuckGo API**, gathering 2400+ images across multiple categories.
- Developed robust **ETL** pipeline handling multiple formats with fallback mechanisms and error recovery.
- Implemented data validation system with multi-stage checks ensuring data quality and consistency.
- Developed a deep learning image classification system using **TensorFlow** and base model **ResNet50**.
- Implemented continuous learning system with user feedback, improving model accuracy over time.

Pokemon Index

Jan 2024 –Feb 2024

- Used **React** to develop a web app to fetch data from **PokeAPI** and display Pokémon details.

AI Story Teller

Nov 2023 – Jan 2024

- Innovatively developed an AI Storyteller app, seamlessly integrating computer vision and **NLP** technologies.
- Demonstrated expertise by leveraging **Hugging Face's Transformers** for image-to-text and integrating **GPT-2** for creative story generation.
- Proficient in **Python**, **ML**, and **deep learning** techniques, with additional capabilities showcased through **OpenAI API** for **NLP** tasks.

AI Story Painter

Nov 2023 – Jan 2024

- Innovatively developed an AI text-to-image app, seamlessly integrating computer vision and **NLP** technologies.
- Demonstrated expertise by leveraging **Hugging Face's Transformers** for image generation and integrating **LLM**.
- Comfortable with different models. Hosted and managed project on **GitHub**, highlighting version control practices.

Search and Rescue using AI/ML

May 2020 – May 2023

- Developed **Raspberry Pi** project to locate and rescue individuals in remote areas with low visibility.
- Developed a program in **C** to fetch real-time **GPS** coordinates and thermal camera images.
- Used Python for SMS communication and **OpenCV** and **Tensorflow** to assist with image processing.
- Developed embedded software that act as a hub for all of the components.

YouTube Clone

May 2022 – Jul 2022

- Developed a video-viewing platform resembling YouTube. Fetched video data (views, likes, channels), enabled search functionality, and categorized videos.
- User Experience: Focus on intuitive interface, efficient content delivery, and user engagement features.

React Live Chat

Mar 2022 – Jun 2022

- Developed a real-time chat application using **Node.js**, **React.js**, and **ChatEngine.io**.
- Implemented user authentication, socket connections, and real-time messaging with features such as group chats, direct messages, and file attachments.
- Troubleshooted and debugged to ensure seamless user experiences, showcasing proficiency in full-stack development and effective problem-solving.

School Database Management

Jan 2020 – Mar 2021

- Developed database with **MySQL** based on the information provided and data cleaning to achieve an organized and easy access database. The project helped over 80 students select their desired GE classes.

Measure Soil Moisture using Microcontroller**Jan 2020 – May 2020**

- Developed embedded software in **C** to interface with **GPIO** for capturing moisture levels in soil.
- Implemented an analog-to-digital converter to display values on an **LCD** screen and make decisions using a finite state machine.

Homework Helper SaaS**Sep 2020 – Dec 2020**

- Developed a Discord bot using **Python** and **Node.js** to assist students with homework and tests.
- Discord server grew to over 200+ students using the service.

Education

Georgia Institute of Technology*M.S. Computer Science***California State University, Sacramento***B.S. Computer Engineering***Additional Information**

Coursera: Machine Learning Stanford University, Mathematics for Machine Learning, Unsupervised Machine Learning, AI & Development by IBM, Relational Database and SQL, Google Data Analytics, Data Visualization with Tableau, Data Science by DeepLearning, Python for Data Science

Award: Dean's Honor, President award, Educational Excellence, Honors Student, Winner of ICPC, Winner of Data Fest Chico

Language: English (Native), Mandarin (Native), Japanese (Elementary)

Citizenship: United State and Taiwan